



Red Hat OpenShift AI Self-Managed 3.0

Managing model registries

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Abstract

As an OpenShift AI administrator, you can create, delete, and manage permissions for model registries in OpenShift AI.

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PREFACE

As an OpenShift AI administrator, you can create, delete, and manage permissions for model registries in OpenShift AI.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog, which includes models from providers such as Red Hat, IBM, Meta, Nvidia, Mistral AI, and Google. OpenShift AI administrators can configure the available repository sources for models displayed in the model catalog.

For more information about how data scientists and AI engineers can use the model catalog, see [Working with the model catalog](#).

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see [Managing model registries](#).

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see [Working with model registries](#).

CHAPTER 2. CREATING A MODEL REGISTRY

You can create a model registry to store, share, version, deploy, and track your models.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see [Enabling the model registry component](#).
- You have access to an external MySQL database which uses at least MySQL version 5.x. However, Red Hat recommends that you use MySQL version 8.x.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Model resources and operations** → **Model registry settings**.
2. Click **Create model registry**.
The **Create model registry** dialog opens.
3. In the **Name** field, enter a name for the model registry.
4. Optional: Click **Edit resource name**, and then enter a specific resource name for the model registry in the **Resource name** field. By default, the resource name will match the name of the model registry.



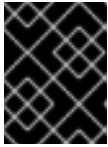
IMPORTANT

Resource names are what your resources are labeled as in OpenShift. Your resource name cannot exceed 253 characters, must consist of lowercase alphanumeric characters or -, and must start and end with an alphanumeric character. Resource names are not editable after creation.

The resource name must not match the name of any other model registry resource in your OpenShift cluster.

5. Optional: In the **Description** field, enter a description for the model registry.
6. In the **Connect to external MySQL database** section, enter the information for the external database where your model data is stored.
 - a. In the **Host** field, enter the database hostname.
 - i. If the database is running in the **rhoai-model-registries** namespace, enter only the hostname for the database.
 - ii. If the database is running in a different namespace from **rhoai-model-registries**, enter the database hostname details in **<hostname>.<namespace>.svc.cluster.local** format.
 - b. In the **Port** field, enter the port number for the database.
 - c. In the **Username** field, enter the default user name that is connected to the database.

- d. In the **Password** field, enter the password for the default user account.
 - e. In the **Database** field, enter the database name.
7. Optional: Select the **Add CA certificate to secure database connection** to use a certificate with your database connection.



IMPORTANT

If your external database is configured to enforce Transport Layer Security (TLS), then you must add a Certificate Authority (CA) certificate.

- a. Click **Use cluster-wide CA bundle** to use the **ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
- b. Click **Use Red Hat OpenShift AI CA bundle** to use the **odh-ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
- c. Click **Choose from existing certificates** to select an existing certificate. You can select the key of any ConfigMap or secret in the **rhoai-model-registries** namespace.
 - i. From the **Resource** list, select a ConfigMap or secret.
 - ii. From the **Key** list, select a key.
- d. Click **Upload new certificate** to upload a new certificate as a ConfigMap.
 - i. Drag and drop the PEM file for your certificate into the **Certificate** field, or click **Upload** to select a file from your local machine's file system.



NOTE

Uploading a certificate creates the **db-credential** ConfigMap with the **ca.crt** key.

To upload a certificate as a secret, you must create a secret in the OpenShift **rhoai-model-registries** namespace, and then select it as an existing certificate when you create your model registry.

For more information about creating secrets, see [Providing sensitive data to pods by using secrets](#) in the OpenShift Container Platform documentation.

8. Click **Create**.



NOTE

To find the resource name or type of a model registry, click the help icon  beside the registry name. Resource names and types are used to find your resources in OpenShift.

Verification

- The new model registry is displayed on the **Model registry settings** page.

- You can edit the model registry by clicking the action menu (⋮) beside it, and then clicking **Edit model registry**.
- You can register a model with the model registry from the **Model registry** tab. For more information about working with model registries, see [Working with model registries](#).

CHAPTER 3. EDITING A MODEL REGISTRY

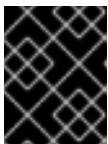
You can edit the details of existing model registries, such as the model registry name, description, and database connection details.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see [Enabling the model registry component](#).
- Your OpenShift AI deployment contains at least 1 model registry.

Procedure

1. From the OpenShift AI dashboard, click **Settings** → **Model resources and operations** → **Model registry settings**.
2. Click the action menu () beside the model registry that you want to edit, and then click **Edit model registry**.
The **Edit model registry** dialog opens.
3. Optional: In the **Name** field, edit the name of the model registry.
4. Optional: In the **Description** field, edit the description of the model registry.
5. Optional: In the **Connect to external MySQL database** section, edit the information for the external database where model data for the registry is stored.
 - a. In the **Host** field, enter the database hostname.
 - i. If the database is running in the **rhoai-model-registries** namespace, enter only the hostname for the database.
 - ii. If the database is running in a different namespace from **rhoai-model-registries**, enter the database hostname details in **<hostname>.<namespace>.svc.cluster.local** format.
 - b. In the **Port** field, enter the port number for the database.
 - c. In the **Username** field, enter the default user name that is connected to the database.
 - d. In the **Password** field, enter the password for the default user account.
 - e. In the **Database** field, enter the database name.
6. Optional: Select the **Add CA certificate to secure database connection** to use a certificate with your database connection.



IMPORTANT

If your external database is configured to enforce Transport Layer Security (TLS), then you must add a Certificate Authority (CA) certificate.

- a. Click **Use cluster-wide CA bundle** to use the **ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
- b. Click **Use Red Hat OpenShift AI CA bundle** to use the **odh-ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
- c. Click **Choose from existing certificates** to select an existing certificate. You can select the key of any ConfigMap or secret in the **rhoai-model-registries** namespace.
 - i. From the **Resource** list, select a ConfigMap or secret.
 - ii. From the **Key** list, select a key.
- d. Click **Upload new certificate** to upload a new certificate as a ConfigMap.
 - i. Drag and drop the PEM file for your certificate into the **Certificate** field, or click **Upload** to select a file from your local machine's file system.



NOTE

Uploading a certificate creates the **db-credential** ConfigMap with the **ca.crt** key.

To upload a certificate as a secret, you must create a secret in the OpenShift **rhoai-model-registries** namespace, and then select it as an existing certificate when you create your model registry.

For more information about creating secrets, see [Providing sensitive data to pods by using secrets](#) in the OpenShift Container Platform documentation.

7. Click **Update**.

Verification

- The model registry is displayed with updated details on the **Model registry settings** page.

CHAPTER 4. MANAGING MODEL REGISTRY PERMISSIONS

You can manage access to a model registry for individual users and user groups in your organization, and for service accounts in a project.

NOTE

OpenShift AI creates the **<model-registry-name>-users** group automatically for use with model registries. You can add users to this group in OpenShift, or ask the cluster administrator to do so.

The model registry operator uses OpenShift Role-Based Access Control (RBAC), and creates various RBAC resources in the **rhoai-model-registries** namespace.

For each model registry instance, the operator creates a **registry-users-<model registry instance name>** role and an OpenShift group called **<model registry instance name>-users**. To grant an individual user, service account, or group access to a model registry instance, your cluster administrator must create a role binding to the **registry-users-<model registry instance name>** role for the instance.

The **<model registry instance name>-users** group has a role binding to the **registry-users-<model registry instance name>** role. Your cluster administrator can add users to this group to grant them access to the model registry instance without needing to create a role binding for each user.

For more information about managing RBAC in OpenShift, see [Using RBAC to define and apply permissions](#).

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- An available model registry exists in your deployment.
- The users and groups that you want to provide access to already exist in OpenShift. For more information, see [Managing users and groups](#).

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations → Model registry settings**.
2. Click **Manage permissions** beside the model registry that you want to manage access for. The permissions page for the model registry opens.
3. Provide one or more OpenShift groups with access to the project.
 - a. On the **Users** tab, in the **Groups** section, click **Add group**.
 - b. From the **Select a group** drop-down list, select a group.

NOTE

To enable access for all cluster users, add **system:authenticated** to the group list.

- c. To confirm your entry, click **Confirm** ().
 - d. Optional: To add an additional group, click **Add group** and repeat the process.
4. Provide one or more users with access to the model registry.
 - a. On the **Users** tab, in the **Users** section, click **Add user**.
 - b. In the **Type username** field, enter the username of the user to whom you want to provide access.
 - c. To confirm your entry, click **Confirm** ().
 - d. Optional: To add an additional user, click **Add user** and repeat the process.
5. Provide all service accounts in a project with access to the model registry.
 - a. On the **Projects** tab, in the **Projects** section, click **Add project**.
 - b. In the **Select or enter a project** field, select or enter the name of the project to which you want to provide access.
 - c. To confirm your entry, click **Confirm** ().
 - d. Optional: To add an additional project, click **Add project** and repeat the process.

Verification

- Users, groups, and accounts that were granted access to a model registry can register, view, edit, version, deploy, delete, archive, and restore models in that registry.
- The **Users** and **Groups** sections on the **Permissions** tab show the respective users and groups that you granted access to the model registry.
- The **Projects** sections on the **Projects** tab show the projects that you granted access to the model registry.

After you provide access to a model registry, users with access can store, share, version, deploy, and track models using the model registry feature. For more information, see [Working with model registries](#).

CHAPTER 5. DELETING A MODEL REGISTRY

You can delete model registries that you no longer require.



IMPORTANT

When you delete a model registry, databases connected to the model registry will not be removed. To remove any remaining databases, contact your cluster administrator.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- An available model registry exists in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations → Model registry settings**.
2. Click the action menu () beside the model registry that you want to delete.
3. Click **Delete model registry**.
4. In the **Delete model registry?** dialog that opens, enter the name of the model registry in the text field to confirm that you intend to delete it.
5. Click **Delete model registry**.

Verification

- The model registry is no longer displayed on the **Model registry settings** page.